

StudyFlix Development Report

Date: 04 September 2026

Module: Assignments Feature (V1 Complete)

✓ Major Accomplishments

✓ 1. Assignment Submission Verification

Confirmed that assignment submissions are successfully written to Firestore.

Firestore Collection

submissions
``

Verified Fields

assignmentId
studentId
submittedAt
status
answers

Submission Example

answers
├ 1 → "34"
├ 2 → "45"
└ 3 → "65"

Result

Student
↓
Submit Assignment
↓
Firestore
↓
Document Created Successfully

✓ 2. Submission Debugging

Added logging to verify Firestore document creation.

Confirmation

Saved document ID:
qRukeGJZZbTnOoQTEQRP

Result

Submission Save Confirmed
Firestore Path Verified

✓ 3. Submission Detection System

Implemented:

```
hasSubmitted(  
  assignmentId,  
  studentId  
)
```

Flow

```
Assignment Opens  
↓  
Check Firestore  
↓  
Has Student Submitted?  
↓  
Return True / False
```

✓ 4. Assignment Completion Tracking

Added:

```
hasSubmitted  
``
```

to UI State.

Result

Assignment Screen

↓

Student Submission Detected

↓

UI Updated Automatically

✓ 5. One Submission Rule

Implemented assignment locking after submission.

Before

Continue To Assignment

↔

always visible.

After

Assignment Submitted

shown instead.

Result

Students cannot accidentally resubmit.

✓ 6. Assignment Status System

Implemented assignment statuses.

Supported States

- Active
- ✓ Submitted
- Closed

Result

Assignment status now updates dynamically based on:

Submission State

+

Due Date

7. Submission Timestamp

Added:

submittedAt

to UI State.

Result

Students can see:

Submitted:

04 Sep 2026 08:42

8. Submitted Assignment Tracking

Implemented:

getSubmittedAssignmentIds()

Result

System identifies all assignments previously submitted by the logged-in student.

9. Assignment List Status Indicators

Assignments screen now displays:

 Submitted

or

 Active

for every assignment card.

Benefit

Students can instantly see completed assignments without opening them.

✓ 10. Assignment Progress Summary

Added progress tracking.

Example

2 of 5 Assignments Submitted

Result

Students now have visibility of assignment completion progress.

✓ 11. Submission Review Feature

Created:

AssignmentSubmissionScreen.kt

Added

View Submission

↓

AssignmentSubmissionScreen

Navigation

Screen.AssignmentSubmission

NavGraph Entry

Route Handling

✓ 12. Submission Data Loading

Extended UI State with:

submission: AssignmentSubmission?

Result

Submission review screen can access:

assignmentId
studentId
submittedAt
answers

✓ 13. Question & Answer Review

Implemented:

Assignment Questions
+
Submitted Answers

Result

Students can now review:

Question 1

Actual Question Text

My Answer:
34

instead of only seeing answer values.

✓ 14. Submission Screen Structure

Added:

Top Bar
Back Button
Submission Summary Card
Assignment Title
Submission Date
Submission Status

Layout

My Submission

Assignment Summary

Question 1

Answer

Question 2

Answer

15. Scrollable Submission Review

Implemented vertical scrolling for long submissions.

Result

Unlimited Question Support

Scrollable Answer Review

16. Teacher Marking Foundation

Expanded `AssignmentSubmission` model.

New Fields

`submissionId`

`assignmentTitle`

`studentName`

`startedAt`

`submittedAt`

`isMarked`

`score`

`totalMarks`

`percentage`

`feedback`

`markedBy`

`markedAt`

Result

Database structure is now ready for:

Teacher Review
Assignment Marking
Feedback
Results
Scoring

without future redesign.

✓ 17. Assignment Timer

Implemented countdown timer.

Flow

Continue To Assignment
↓
Timer Starts
↓
Counts Down Every Second

Example

01:00:00
00:59:59
00:59:58
...

Result

Assignments now support timed assessments.

✓ 18. Time Expiry Detection

Implemented:

Timer Reaches 00:00:00
↓

Time Expired Event Triggered

..

Result

Foundation prepared for automatic time-based submission.

19. Submission Confirmation Dialog

Added confirmation dialog before final submission.

New Flow

Submit Assignment

↓

Are You Sure?

↓

Cancel / Submit

Benefit

Prevents accidental submissions.

20. Overdue Assignment Detection

Implemented due-date validation using:

dueDate

Supported Logic

Active Assignment

↓

Can Open

Due Date Passed

↓

Closed

Result

Students cannot access expired assignments.

21. Closed Assignment Locking

Before

Status: Active

[Continue To Assignment]

After

Status: Closed 

[Assignment Closed]

Result

Overdue assignments are automatically locked.

Current Assignment Feature Status

Fully Working

Assignment List

Assignment Details

Instructions

Question Screen

Answer Entry

Countdown Timer

Confirmation Dialog

Firebase Submission

Submission Detection

Submission Tracking

Submission Date

Submission Review

Question Review

Assignment Progress Summary

Submitted Assignment Indicators

Assignment Status

Active Status

Submitted Status

Closed Status

Overdue Detection

Closed Assignment Locking

Teacher-ready Submission Metadata

Navigation

Back Navigation



Assignment V1 Status

COMPLETE

Student Workflow

Open Assignment



Read Instructions



Start Timed Assessment



Answer Questions



Confirm Submission



Save To Firestore



Track Submission



Review Submission



Planned Assignment V2 Features

Student Side

Multiple Attempts

Highest Score Tracking
Draft Saving
Persistent Timer Across Sessions
Memo Viewing
Results Screen
Teacher Feedback Display
Assignment Analytics

Teacher Side

Submitted Assignments
Marking Dashboard
Marks Allocation
Feedback System
Result Publishing
Memo Management

Final Morning / Afternoon Milestone

Student
↓
Open Assignment
↓
Start Timer
↓
Answer Questions
↓
Confirm Submission
↓
Firestore Save
↓
Submission Tracked
↓
Status Updated
↓
Assignment Locked
↓
Review Submission

Git Commit  Assignment Module = V1 Complete  

